



MATHEMATICS (EXTENSION 1)

2012 HSC Course Assessment Task 4

August 15, 2012

General instructions

- Working time – 50 min.
- **Commence each new question on a new page.**
- Write using blue or black pen. Where diagrams are to be sketched, these may be done in pencil.
- Board approved calculators may be used.
- All necessary working should be shown in every question. Marks may be deducted for illegible or incomplete working.
- Attempt **all** questions.
- At the conclusion of the examination, bundle the booklets used in the correct order within this paper and hand to examination supervisors.

Class (please ✓)

- ☐ 12M3C – Ms Ziazaris
- ☐ 12M3D – Mr Lowe
- ☐ 12M3E – Mr Lam
- ☐ 12M4A – Mr Lin
- ☐ 12M4B – Mr Ireland
- ☐ 12M4C – Mr Fletcher

STUDENT NUMBER # BOOKLETS USED:

Marker's use only.

QUESTION	1	2	3	4	5	Total	%
MARKS	$\overline{8}$	$\overline{7}$	$\overline{9}$	$\overline{7}$	$\overline{8}$	$\overline{39}$	

Question 1 (8 Marks)

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Marks

(a) Prove $\frac{d}{dx} \left(\frac{1}{2} v^2 \right) = \frac{d^2 x}{dt^2}$. **2**

(b) The velocity of a particle moving along the x axis is given by

$$v^2 = 4(1 + x^2)^2$$

Initially, the particle is at $x = 1$ with velocity $v = 4 \text{ ms}^{-1}$.

i. Find value of the initial acceleration. **2**

ii. Show the displacement of the particle as a function of time is **3**

$$x = \tan \left(2t + \frac{\pi}{4} \right)$$

iii. Explain why $0 \leq t < \frac{\pi}{8}$ for this motion to be valid. **1**

Question 2 (7 Marks)

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Marks

(a) Consider the expansion of $(1 + 2x)^n$.

i. Write down an expression for the coefficient of the term in x^4 . **2**

ii. The ratio of the coefficient of the term in x^4 to that of the term in x^6 is $5 : 8$. Find the value of n . **3**

(b) Find the coefficient of x^3 in the expansion of $(1 - 2x)^{18}(1 + 3x)^{17}$. **2**

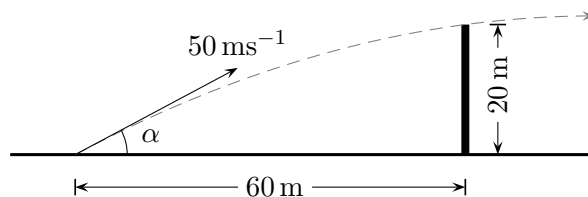
Question 3 (9 Marks)

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Marks

A particle is projected from ground level with an initial speed of 50 ms^{-1} towards a wall which is 60 m from the point of projection, and 20 m high and just scrapes past the wall (i.e. does not collide with it).

Take $g = 10 \text{ ms}^{-2}$.



(a) Show that the equations of motion for this system are **3**

$$\begin{cases} x = 50t \cos \alpha \\ y = -5t^2 + 50t \sin \alpha \end{cases}$$

(b) Show that $9 \tan^2 \alpha - 75 \tan \alpha + 34 = 0$. **3**

(c) Hence or otherwise, find the angle(s) of projection for which the particle will just scrape past the wall, correct to the nearest degree. **3**

Question 4 (7 Marks)

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Marks

Samuel Sung opened a warehouse on 1 July 2012, distributing *SolarSystem S3* mobile phones to retailers.

His initial stock was 10 000 units of the phone. During any month, he sells 25% of the existing stock at the beginning of that month. In order to keep up with demand, he purchases an additional 100 phones on the last day of each month.

- (a) Show that the number of phones in the warehouse at the end of the second month is **2**

$$A_2 = 10\,000 \times 0.75^2 + 100(1 + 0.75)$$

- (b) Show that A_n , the number of *SolarSystem S3* phones in stock after n months is given by **3**

$$A_n = 9\,600 \times 0.75^n + 400$$

- (c) After how many months of opening the warehouse will Samuel distribute less than 500 phones to retailers? **2**

Question 5 (8 Marks)

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Marks

- (a) By considering the expansion of $(1+x)^n$, show that **3**

$$2\binom{n}{2} + 6\binom{n}{3} + 12\binom{n}{4} + \cdots + n(n-1)\binom{n}{n} = n(n-1)2^{n-2}$$

- (b) i. State the number of terms in this geometric series: **1**

$$1 + (1+x) + (1+x)^2 + (1+x)^3 + \cdots + (1+x)^n$$

- ii. Express $1 + (1+x) + (1+x)^2 + (1+x)^3 + \cdots + (1+x)^n$ in simplest terms, using the formula for the sum of a geometric progression. **2**

- iii. By considering the coefficient of x^r , where $0 < r \leq n$, in the expansion of **2**

$$1 + (1+x) + (1+x)^2 + (1+x)^3 + \cdots + (1+x)^n$$

Prove that

$$\binom{n}{r} + \binom{n-1}{r} + \binom{n-2}{r} + \cdots + \binom{r}{r} = \binom{n+1}{r+1}$$

End of paper.

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1} + C, \quad n \neq -1; \quad x \neq 0 \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x + C, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax} + C, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax + C, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax + C, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax + C, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax + C, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a} + C, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a} + C, \quad a > 0, -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right) + C, \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right) + C$$

NOTE: $\ln x = \log_e x$, $x > 0$

Suggested Solutions

Question 1 (Ziaziaris)

(a) (2 marks)

$$\begin{aligned}\frac{d}{dx} \left(\frac{1}{2} v^2 \right) &= \frac{d}{dv} \left(\frac{1}{2} v^2 \right) \frac{dv}{dx} \quad (\text{chain rule}) \\ &= v \frac{dv}{dx} \\ &= \frac{dx}{dt} \cdot \frac{dv}{dx} \\ &= \frac{dv}{dt} = \frac{d^2 x}{dt^2} \\ &= a\end{aligned}$$

(b) i. (2 marks)

✓ [1] for differentiating to obtain

$$a = 8x(1 + x^2)$$

✓ [1] obtaining $a = 8 \text{ ms}^{-2}$ after substitution and evaluation.

$$\begin{aligned}v^2 &= 4(1 + x^2)^2 \\ \frac{d}{dx} \left(\frac{1}{2} v^2 \right) &= \frac{d}{dx} \left(2(1 + x^2)^2 \right) \\ &= 2 \cdot 2x \cdot 2 \cdot (1 + x^2) \\ &= 8x(1 + x^2) \Big|_{x=1} \\ &= 16 \text{ ms}^{-2}\end{aligned}$$

ii. (3 marks)

✓ [1] for obtaining $\frac{dx}{dt} = 2(1 + x^2)$.

✓ [1] for obtaining $\tan^{-1} x = 2t + C$.

✓ [1] for value of C .

$$v^2 = 4(1 + x^2)^2$$

When $x = 1$, $v > 0$. Hence use positive square root.

$$\begin{aligned}v &= \frac{dx}{dt} = 2(1 + x^2) \\ \frac{dx}{1 + x^2} &= 2 dt\end{aligned}$$

Integrating,

$$\tan^{-1} x = 2t + C$$

When $t = 0$, $x = 1$.

$$\begin{aligned}\tan^{-1} 1 &= \frac{\pi}{4} = 0 + C \\ \therefore C &= \frac{\pi}{4} \\ \therefore x &= \tan \left(2t + \frac{\pi}{4} \right)\end{aligned}$$

iii. (1 mark)

✓ [1] awarded only for fully justified answers.

As it is not physically possible to cross $t = \frac{\pi}{2}$ when $x = \tan t$ (which makes particle “teleport” from $x = \infty$ to $x = -\infty$!), inspect the domain of $\tan t$ in its first period only:

$$D = \left\{ t : -\frac{\pi}{2} < t < \frac{\pi}{2} \right\}$$

Inspecting the domain of $\tan \left(2t + \frac{\pi}{4} \right)$:

$$\begin{aligned}D &= \left\{ t : -\frac{\pi}{2} < 2t + \frac{\pi}{4} < \frac{\pi}{2} \right\} \\ &= \left\{ t : -\frac{3\pi}{4} < 2t < \frac{\pi}{4} \right\}\end{aligned}$$

As $t > 0$,

$$D = \left\{ t : 0 \leq 2t < \frac{\pi}{4} \right\} = \left\{ t : 0 \leq t < \frac{\pi}{8} \right\}$$

Question 2 (Lowe)

(a) i. (2 marks)

✓ [-1] if the term (rather than coefficient only) is given.

$$(1 + 2x)^n = \sum_{k=0}^n \binom{n}{k} 2^k x^k$$

Coefficient of term in x^4 :

$$\binom{n}{4} 2^4$$

ii. (3 marks)

✓ [1] for simplifying ratio of coefficient of term in x^4 to term in x^6 .

✓ [1] for obtaining quadratic after equating with ratio of $\frac{5}{8}$.

✓ [1] for justifying $n = 8$ only.

The term in x^6 is

$$\binom{n}{6} 2^6$$

Ratio of term in x^4 to that in x^6 :

$$\begin{aligned} \frac{\binom{n}{4} \times 2^4}{\binom{n}{6} \times 2^6} &= \frac{\cancel{n!} 2^4}{\cancel{n!} (n-4)(n-5)(n-6)!} \\ &= \frac{2^4}{6 \times 5 \times \cancel{n!} (n-6)!} \times 2^2 \\ &= \frac{30}{4(n-4)(n-5)} \\ &= \frac{15}{2(n-4)(n-5)} \end{aligned}$$

The terms are in the ratio 5 : 8:

$$\begin{aligned} \frac{15^3}{2(n-4)(n-5)} &= \frac{8}{4} \\ 12 &= (n-4)(n-5) \\ n^2 - 9n + 20 &= 12 \\ n^2 - 9n + 8 &= 0 \\ (n-8)(n-1) &= 0 \\ \therefore n &= 8 \text{ only} \end{aligned}$$

as $n = 1$ produces no x^4 or x^6 term.

(b) (2 marks)

✓ [1] for pairs of k and r that add to required index.

✓ [1] for final answer.

$$\begin{aligned} &(1-2x)^{18}(1+3x)^{17} \\ &= \left(\sum_{k=0}^{18} \binom{18}{k} (-1)^k 2^k x^k \right) \left(\sum_{r=0}^{17} \binom{17}{r} 3^r x^r \right) \end{aligned}$$

The typical term in this expansion is

$$\binom{18}{k} \binom{17}{r} (-1)^k 2^k 3^r x^{k+r}$$

The coefficient of x^3 appears when $k+r=3$:

k	r	$k+r$	Coefficient
0	3	3	$\binom{18}{0} \binom{17}{3} 2^0 3^3$
1	2	3	$-\binom{18}{1} \binom{17}{2} 2^1 3^2$
2	1	3	$\binom{18}{2} \binom{17}{1} 2^2 3^1$
3	0	3	$-\binom{18}{3} \binom{17}{0} 2^3 3^0$

Hence the coefficient of the term in x^3 is

$$\begin{aligned} &\binom{18}{0} \binom{17}{3} 2^0 3^3 - \binom{18}{1} \binom{17}{2} 2^1 3^2 \\ &+ \binom{18}{2} \binom{17}{1} 2^2 3^1 - \binom{18}{3} \binom{17}{0} 2^3 3^0 \end{aligned}$$

(Stop at this point and ignore any further working to simplify)

Question 3 (Fletcher)

(a) (3 marks)

✓ [1] for correct basis equations

$$\ddot{x} = 0 \quad \ddot{y} = -g$$

✓ [1] for each correct derivation of x and y in terms of t .

$$\begin{cases} \ddot{x} = 0 \\ \ddot{y} = -g \end{cases}$$

Finding x by integrating twice:

$$\dot{x} = \int \ddot{x} dt = \int 0 dt = C_1$$

When $t = 0$, the component of velocity is

$$\begin{aligned} \dot{x} &= 50 \cos \alpha \\ x &= \int \dot{x} dt = \int 50 \cos \alpha dt \\ &= 50t \cos \alpha + C_2 \end{aligned}$$

When $t = 0$, $x = 0$. Hence $C_2 = 0$.

$$\therefore x = 50t \cos \alpha$$

Finding y by integrating twice:

$$\dot{y} = \int \ddot{y} dt = \int -g dt = -gt + C_3$$

When $t = 0$, the component of velocity is

$$\begin{aligned}\dot{y} &= -g(0) + C_3 = 50 \sin \alpha \\ \therefore C_3 &= 50 \sin \alpha \\ \therefore \dot{y} &= -gt + 50 \sin \alpha \\ y &= \int \dot{y} dt \\ &= \int -gt + 50 \sin \alpha dt \\ &= -\frac{1}{2}gt^2 + 50t \sin \alpha + C_4\end{aligned}$$

When $t = 0$, $y = 0$. Hence $C_4 = 0$.

$$\therefore y = -\frac{1}{2}gt^2 + 50t \sin \alpha$$

(b) (3 marks)

- ✓ [1] for Cartesian equation, eliminating t .
- ✓ [1] for substituting $x = 60$, $y = 20$.
- ✓ [1] for final answer.

$$\begin{cases} y = -5t^2 + 50t \sin \alpha \\ x = 50t \cos \alpha \end{cases} \\ \therefore t = \frac{x}{50 \cos \alpha}$$

Substitute to $y = -5t^2 \dots$:

$$\begin{aligned}y &= -5 \left(\frac{x}{50 \cos \alpha} \right)^2 + 50 \left(\frac{x}{50 \cos \alpha} \right) \sin \alpha \\ &= -\frac{x^2}{500 \cos^2 \alpha} + x \tan \alpha \\ &= -\frac{x^2}{500} \sec^2 \alpha + x \tan \alpha \\ &= -\frac{x^2}{500} (1 + \tan^2 \alpha) + x \tan \alpha\end{aligned}$$

When particle scrapes past, $x = 60$ and $y = 20$.

$$\begin{aligned}\underbrace{20}_{\times 5} &= -\underbrace{\frac{6 \times 6 \times 10^2}{5 \times 10^2}}_{\times 5} (1 + \tan^2 \alpha) + \underbrace{60}_{\times 5} \tan \alpha \\ 25 &= -9 - 9 \tan^2 \alpha + 75 \tan \alpha \\ \therefore 9 \tan^2 \alpha - 75 \tan \alpha + 34 &= 0\end{aligned}$$

(c) (3 marks)

- ✓ [1] for $\tan \alpha = \frac{25 \pm \sqrt{489}}{6}$ (or unsimplified equivalent)
- ✓ [1] each for $\alpha = 26^\circ$ or $\alpha = 83^\circ$

Let $m = \tan \alpha$.

$$\begin{aligned}9m^2 - 75m + 34 &= 0 \\ m &= \frac{75 \pm \sqrt{75^2 - 4(9)(34)}}{2 \times 9} \\ &= \frac{75 \pm 3\sqrt{489}}{18} \\ &= \frac{25 \pm \sqrt{489}}{6} \\ \therefore \tan \alpha &= \frac{25 \pm \sqrt{489}}{6} \\ \alpha &= 26^\circ, 83^\circ\end{aligned}$$

Question 4 (Lin)

(a) (2 marks)

- ✓ [1] for A_1 .
- ✓ [1] for A_2 .

$$\begin{aligned}A_1 &= 10\,000 \times 0.75 + 100 \\ A_2 &= A_1 \times 0.75 + 100 \\ &= (10\,000 \times 0.75 + 100) \times 0.75 \\ &\quad + 100 \\ &= 10\,000 \times 0.75^2 + 100(1 + 0.75)\end{aligned}$$

(b) (3 marks)

- ✓ [1] for correctly generalising to A_n .
- ✓ [1] for $S_n = 4(1 - 0.75^n)$.
- ✓ [1] for final answer.

$$\begin{aligned}A_n &= 10\,000 \times 0.75^n \\ &\quad + 100 \underbrace{(1 + 0.75 + \dots + 0.75^{n-1})}_{\text{GP: } a=1, r=0.75} \\ S_n &= \frac{1(1 - 0.75^n)}{1 - 0.75} = \frac{(1 - 0.75^n)}{\frac{1}{4}} \\ &= 4(1 - 0.75^n) \\ \therefore A_n &= 10\,000 \times 0.75^n + 400(1 - 0.75^n) \\ &= 0.75^n (10\,000 - 400) + 400 \\ &= 9\,600 \times 0.75^n + 400\end{aligned}$$

(c) (2 marks)

- ✓ [1] for $A_n \times 0.25 < 500$.
- ✓ [1] for final answer.
- The number of phones distributed is $A_n \times 0.25$ (distributes 25% of what is in the warehouse).

$$(9\,600 \times 0.75^n + 400) \times 0.25 < 500$$

$$9\,600 \times 0.75^n + 400 < \frac{500}{0.25} = 2000$$

$$\frac{9\,600}{\div 9\,600} \times 0.75^n < \frac{1\,600}{\div 9\,600}$$

$$0.75^n < \frac{1}{6}$$

$$n \log 0.75 < \log \frac{1}{6}$$

$$n > \frac{\log \frac{1}{6}}{\log 0.75} = 6.228 \dots$$

Less than 500 phones will be distributed after 7 months.

Question 5 (Ireland)

(a) (3 marks)

- ✓ [1] for correctly differentiating each time.
- ✓ [1] for substituting $x = 1$ and obtaining correct expression.

$$(1+x)^n = \sum_{k=0}^n \binom{n}{k} x^k$$

Differentiating,

$$n(1+x)^{n-1} = \binom{n}{1} + 2\binom{n}{2}x + \dots + n\binom{n}{n}x^{n-1}$$

Differentiate again,

$$\begin{aligned} & n(n-1)(1+x)^{n-2} \\ &= 2\binom{n}{2} + 3 \cdot 2\binom{n}{3}x + \dots + n(n-1)\binom{n}{n}x^{n-2} \end{aligned}$$

When $x = 1$,

$$n(n-1)2^{n-2} = 2\binom{n}{2} + 6\binom{n}{3} + \dots + n(n-1)\binom{n}{n}$$

(b) i. (1 mark) $n + 1$.

ii. (2 marks)

- ✓ [1] for correct substitution into sum of GP formula.
- ✓ [1] for final answer.

$$1 + (1+x) + (1+x)^2 + (1+x)^3 + \dots + (1+x)^n$$

GP: $a = 1$, $r = (1+x)$.

$$\begin{aligned} S_{n+1} &= \frac{1((1+x)^{n+1} - 1)}{(1+x) - 1} \\ &= \frac{1}{x}((1+x)^{n+1} - 1) \end{aligned}$$

iii. (2 marks)

- ✓ [1] for finding coef in x^r as $\binom{n}{r} + \binom{n-1}{r} + \dots + \binom{n}{r}$.
- ✓ [1] for obtaining coef of x^{r+1} which is $\binom{n+1}{r+1}$.

$$\begin{aligned} & \frac{1}{x}((1+x)^{n+1} - 1) \\ &= 1 + (1+x) + (1+x)^2 + \dots + (1+x)^n \end{aligned}$$

Examine the coefficient of x^r in $\frac{1}{x}((1+x)^{n+1} - 1)$:

- x^r term appears when the power is $\geq r$. Relevant terms are:

$$\binom{r}{r} \quad \binom{r+1}{r} \quad \binom{r+2}{r} \dots \binom{n}{r}$$

In $1 + (1+x) + (1+x)^2 + \dots + (1+x)^n$, the term required is x^{r+1} as x^{-1} exists to reduce the index by 1. The coefficient is thus

$$\binom{n+1}{r+1}$$

Equating coefficients in x^r ,

$$\begin{aligned} \binom{r}{r} + \binom{r+1}{r} + \dots + \binom{n-1}{r} + \binom{n}{r} \\ = \binom{n+r}{r+1} \end{aligned}$$